

INDUSTRIAL WATER & WASTEWATER ENGINEERING SERIES

# EFFLUENT TREATMENT PLANT (ETP)

Detailed Engineering & Operations Manual — Expanded Edition

DETAILED ENGINEERING & OPERATIONS MANUAL — EXPANDED EDITION

Water Chemistry Deep-Dives · Process Flow Diagrams · Design, O&M and Project Reference

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Industrial Water & Wastewater Engineering Series · Reference Document for Design, O&M and Project Teams

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## 1. Plant Profiling & Overview

An Effluent Treatment Plant (ETP) is a dedicated industrial installation designed to treat process wastewater generated from manufacturing operations before it is discharged to a municipal sewer, water body, or reused within the plant. Unlike a Sewage Treatment Plant (STP), which primarily handles domestic-strength organic wastewater, an ETP is engineered around the specific pollutant load of an industry – which may include high COD/BOD, oils and grease, heavy metals, dyes, suspended solids, dissolved salts, or toxic/recalcitrant organics.

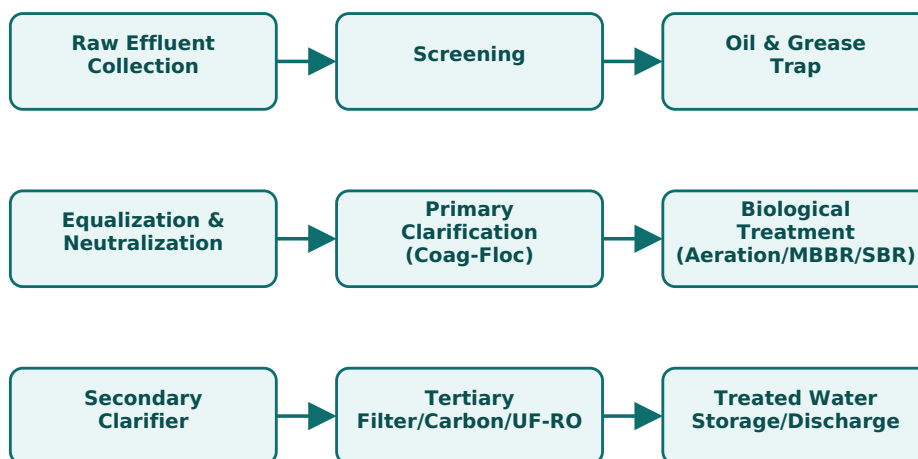


Fig. 1 — ETP Process Train (mainstream), with parallel sludge line



Fig. 2 — Parallel Sludge Handling Line

### 1.1 Objectives of an ETP

- Reduce pollutant concentration (BOD, COD, TSS, oil & grease, heavy metals, color) to levels prescribed by the Pollution Control Board (PCB) / CPCB discharge norms.
- Enable safe discharge into inland surface water, sewer, or on land for irrigation.
- Enable water recovery and recycle to reduce fresh-water intake (increasingly mandated under Zero Liquid Discharge policies for red/orange category industries).
- Protect downstream infrastructure (sewers, CETPs) and receiving ecosystems from toxic shock loads.
- Recover valuable by-products where feasible (oil skimming, metal recovery, salt recovery in ZLD).

### 1.2 Typical Applicability

ETPs are mandatory for pharmaceutical, textile & dyeing, tannery, pulp & paper, chemical, petrochemical, pesticide, distillery, dairy, food processing, electroplating, automobile, and pigment/dye manufacturing units. Capacity ranges from a few KLD (kilolitres per day) for small scale units to several MLD (million litres per day) for large integrated complexes.

### 1.3 Regulatory Framework (India context, adaptable globally)

- Water (Prevention & Control of Pollution) Act, 1974 – Consent to Establish (CTE) and Consent to Operate (CTO) from State PCB.
- Environment (Protection) Act, 1986 – discharge standards under Schedule VI.
- CPCB/SPCB category-wise ZLD mandates for textile, tannery, distillery and pulp & paper sectors.
- Hazardous & Other Wastes (Management) Rules for ETP sludge disposal.

## 2. Industry-wise Effluent Characterization

Design of the ETP begins with a detailed characterization of the raw effluent since unit processes, chemical dosing, and detention times are all a function of pollutant strength. Typical ranges are summarized below; actual plant design must be based on a minimum of 3–6 months of representative sampling (diurnal and seasonal variation).

| Industry                | Key Pollutants                         | BOD (mg/L)  | COD (mg/L)   | TSS (mg/L)  | Special Concerns                                   |
|-------------------------|----------------------------------------|-------------|--------------|-------------|----------------------------------------------------|
| Textile & Dyeing        | Color, dyes, salts, alkalinity         | 150-600     | 800-3000     | 100-500     | High TDS, color removal, ZLD mandate               |
| Tannery                 | Chromium, sulfide, high TDS            | 1000-4000   | 3000-10000   | 2000-6000   | Cr(VI) reduction, odor, sludge toxicity            |
| Pharmaceutical          | COD, solvents, antibiotics (biocidal)  | 500-5000    | 2000-15000   | 200-1000    | Bio-inhibition, advanced oxidation needed          |
| Pulp & Paper            | Lignin, color, AOX                     | 300-1200    | 1000-4000    | 500-2000    | Color & AOX removal, large volumes                 |
| Distillery (Spent Wash) | Very high organic load, dark color     | 40000-60000 | 80000-160000 | 10000-30000 | Anaerobic pre-treatment (biomethanation) essential |
| Electroplating          | Heavy metals (Cr, Ni, Cu, Zn), cyanide | 50-200      | 200-800      | 50-300      | Metal precipitation, cyanide destruction           |
| Dairy/Food              | Fats, oil, organics                    | 1000-4000   | 2000-8000    | 500-1500    | Grease trap, high biodegradability                 |
| Petrochemical           | Oil, phenols, sulfides                 | 200-800     | 500-2500     | 100-400     | Oil-water separation, phenol removal               |

Note: Values are indicative ranges compiled from common industry practice and vary widely by process, product mix and water-use intensity. Site-specific characterization is mandatory before finalizing design.

## 3. Process Design & Engineering Basis

### 3.1 Design Basis Parameters

- Design flow (average and peak) – derived from production capacity and water balance, with 20-25% safety margin for peak/shock flows.
- Pollutant loading (kg BOD/COD per day) used to size biological reactors via F/M ratio (Food-to-Microorganism ratio, typically 0.2-0.4 kg BOD/kg MLVSS.day for activated sludge).
- Hydraulic Retention Time (HRT) for each unit – equalization 6-12 hr, aeration 12-24 hr (higher for high-strength industrial effluent), clarifiers 2-4 hr.
- Mixed Liquor Suspended Solids (MLSS) maintained typically 2500-4500 mg/L in conventional activated sludge; up to 6000-8000 mg/L in MBBR/MBR systems.
- Sludge age (Solids Retention Time, SRT) 8-20 days depending on nitrification requirement.
- Oxygen transfer requirement computed from BOD load, endogenous respiration and nitrification demand; blower capacity sized with 1.5x safety factor.

### 3.2 Core Unit Operations

**a) Equalization Tank:** Dampens hydraulic and organic shock loads from batch discharges; sized for 6-12 hours of average flow with mechanical/air-agitation to prevent settling.

**b) Coagulation-Flocculation (Primary Treatment):** Chemical dosing (alum, ferric chloride/ferrous sulphate, poly-aluminium chloride, and polyelectrolyte) to destabilize colloidal solids, followed by flash mixing (60-90 sec) and slow flocculation (15-20 min) ahead of a primary clarifier or Dissolved Air Flotation (DAF) unit.

**c) Biological Treatment** selected based on effluent strength and land availability: Activated Sludge Process (ASP); Extended Aeration; Moving Bed Biofilm Reactor (MBBR); Sequential Batch Reactor (SBR); Membrane Bioreactor (MBR); Anaerobic pretreatment (UASB/Biomethanation) for very high COD streams, reducing downstream aerobic load by 70-85%.

**d) Secondary Clarifier:** Gravity separation of biomass from treated water; surface overflow rate typically 16-24 m<sup>3</sup>/m<sup>2</sup>/day; sludge recycled to aeration tank (RAS) and excess wasted (WAS) to sludge handling.

**e) Tertiary/Polishing Treatment:** Pressure sand filter → activated carbon filter → (optional) ultrafiltration → Reverse Osmosis, required when treated water is to be reused for cooling, gardening, or process makeup.

## 4. Pretreatment Systems

| Stage | Purpose |
|-------|---------|
|-------|---------|

| Typical Equipment             |                                     |                                                                          | Design Notes                                   |
|-------------------------------|-------------------------------------|--------------------------------------------------------------------------|------------------------------------------------|
| Bar Screen (Coarse)           | Remove large solids, rags, plastics | Manual/mechanical bar screen, 20-40mm spacing                            | Prevents pump/pipe clogging                    |
| Fine Screen                   | Remove fibrous & fine solids        | Static wedge-wire or rotary drum screen, 1-6mm                           | Protects downstream pumps                      |
| Oil & Grease Trap / DAF       | Remove free & emulsified oil        | API separator, Tilted Plate Interceptor, DAF with air dissolution        | Essential for petrochemical, dairy, auto units |
| Equalization & Neutralization | Buffer flow/load, adjust pH 6.5-8.5 | RCC/steel tank with diffused air or mechanical mixer, acid/alkali dosing | pH probes with auto-dosing interlock           |
| Grit Chamber                  | Remove sand/grit (where applicable) | Vortex or horizontal flow grit chamber                                   | Common in combined effluent/CETP streams       |

## 5. Primary, Secondary & Tertiary Treatment

The three-stage treatment train progressively removes suspended, colloidal, and dissolved pollutants. Primary treatment (physico-chemical) targets 60-70% BOD and 80-90% TSS removal. Secondary (biological) treatment further reduces BOD/COD by 85-95% overall. Tertiary treatment polishes the effluent to below-discharge or reuse-grade quality (turbidity <5 NTU, BOD <10 mg/L, COD <50 mg/L for recycling applications).

Advanced Oxidation Processes (AOP – ozone/UV/H<sub>2</sub>O<sub>2</sub>/Fenton) and activated carbon adsorption are increasingly used for recalcitrant COD, color and micro-pollutants that resist biological degradation, particularly in pharmaceutical and dye-intermediate effluents.

## 6. Sludge Handling & Zero-Waste Disposal

- **Thickening:** Gravity thickener or mechanical thickener concentrates sludge from 0.5-1% to 3-5% solids before dewatering.
- **Dewatering:** Filter press (chamber/membrane plate), decanter centrifuge, or belt press reduces moisture to 65-75% cake, aided by polyelectrolyte conditioning.
- **Stabilization:** Aerobic/anaerobic digestion or lime stabilization reduces pathogens and odor before disposal.
- **Disposal Routes:** Secured landfill (for hazardous sludge, e.g., tannery chrome sludge, electroplating metal sludge), co-processing in cement kilns, or land application (non-hazardous, nutrient-rich sludge) as per SPCB authorization.
- **ZLD Linkage:** Where full Zero Liquid Discharge is mandated, treated effluent from tertiary stage is routed to RO → RO reject to evaporator/crystallizer, recovering both water and solid salts (see the dedicated ZLD manual).

## 7. Effluent & Biological Process Chemistry Deep-Dive

### 7.1 Coagulation Chemistry

Alum:  $\text{Al}_2(\text{SO}_4)_3 \cdot 14\text{H}_2\text{O} + 6\text{HCO}_3^- \rightarrow 2\text{Al}(\text{OH})_3(\text{s}) + 3\text{SO}_4^{2-} + 6\text{CO}_2 + 14\text{H}_2\text{O}$   
Ferric chloride:  $\text{FeCl}_3 + 3\text{HCO}_3^- \rightarrow \text{Fe}(\text{OH})_3(\text{s}) + 3\text{Cl}^- + 3\text{CO}_2$

Coagulants hydrolyze to form positively charged metal-hydroxide species that neutralize the negative surface charge on colloidal particles (charge neutralization) and subsequently form a sweep-floc of  $\text{Al}(\text{OH})_3/\text{Fe}(\text{OH})_3$  that entraps fine solids. Optimum dose is determined by jar testing and is strongly pH-dependent – alum performs best at pH 6.0-7.5, ferric salts over a wider pH 5.5-8.5 range. Coagulant demand consumes alkalinity, so low-alkalinity effluents may need supplemental lime/soda ash to hold pH in the effective coagulation band.

### 7.2 F/M Ratio & Biological Kinetics (worked example)

$$\text{F/M} = \text{BOD load (kg/day)} / (\text{MLVSS (kg/m}^3) \times \text{Aeration tank volume (m}^3))$$

Example: BOD load = 500 kg/day; target F/M = 0.3; MLVSS = 3.0 kg/m<sup>3</sup>

Required MLVSS mass =  $500 / 0.3 = 1667 \text{ kg}$

Aeration volume =  $1667 / 3.0 = 556 \text{ m}^3$

Check HRT at flow 2000 m<sup>3</sup>/day:  $\text{HRT} = 556/2000 \times 24 = 6.7 \text{ hr}$  (increase volume if higher HRT is targeted for high-strength effluent)

A low F/M (extended aeration, F/M 0.05-0.15) favors a stable, low-yield, well-settling biomass; a higher F/M (conventional ASP, 0.2-0.4) gives a more compact reactor but is more sensitive to shock loads and produces more sludge – this trade-off is the basis for the reactor-type selection in Section 3.2.

### 7.3 Sludge Volume Index (SVI) & Bulking Diagnosis

$$\text{SVI (mL/g)} = (\text{Settled sludge volume after 30 min, mL/L}) / (\text{MLSS, g/L})$$

SVI < 100 → good settling floc

SVI 100-200 → moderate, monitor

SVI > 200 → filamentous bulking likely – check DO, F/M, nutrient balance

Filamentous bulking is commonly triggered by low dissolved oxygen (<1-2 mg/L), nutrient deficiency (C:N:P imbalance, target ratio 100:5:1), or a very low F/M/high sludge age – the corrective direction depends on which of these is out of range, which is why SVI trending is paired with DO and nutrient testing rather than treated alone.

### 7.4 Chromium Reduction & Metal Precipitation (electroplating/tannery)

Cr(VI) reduction:  $2\text{CrO}_4^{2-} + 3\text{SO}_2 + 4\text{H}^+ \rightarrow \text{Cr}_2(\text{SO}_4)_3\text{-type species (Cr}^{3+}) + \dots$  (typically via SO<sub>2</sub>/sodium bisulfite at pH 2-3)

Metal hydroxide precipitation:  $\text{M}^{2+} + 2\text{OH}^- \rightarrow \text{M}(\text{OH})_2(\text{s})$  [M = Cr<sup>3+</sup>, Ni<sup>2+</sup>, Cu<sup>2+</sup>, Zn<sup>2+</sup>]

Hexavalent chromium (Cr<sup>6+</sup>, carcinogenic and soluble) must first be chemically reduced to trivalent chromium (Cr<sup>3+</sup>) under acidic conditions before it can be precipitated as Cr(OH)<sub>3</sub> by raising pH to 8.5-9.5 with lime/caustic. Each metal hydroxide has a different minimum-solubility pH, so multi-metal streams are often precipitated in two pH-controlled stages to achieve compliance on all metals simultaneously.

### 7.5 Nutrient Balance (C:N:P)

Biological treatment requires nitrogen and phosphorus in addition to carbon for cell synthesis, at an approximate BOD:N:P ratio of 100:5:1. Many industrial effluents (distillery, dairy) are nitrogen/phosphorus deficient relative to their carbon load and require supplemental urea/DAP dosing; conversely, effluents with excess nutrients (dairy, food processing) may need denitrification if discharge norms include a total nitrogen limit.

## 8. Common Problems & Engineering Solutions

| Problem | Root Cause | Engineering Solution |
|---------|------------|----------------------|
|---------|------------|----------------------|

|                                                    |                                                                       |                                                                                                         |
|----------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Sludge bulking (poor settling, filamentous growth) | Low F/M ratio, nutrient deficiency, low DO, septic feed               | Optimize F/M via load balancing, add nutrients (N:P), maintain DO 2-3 mg/L, chlorinate RAS if severe    |
| Foaming in aeration tank                           | Nocardia/filamentous bacteria, detergents, high MLSS age              | Reduce sludge age, spray water/anti-foam, skim and waste foam-laden sludge                              |
| Odor around plant                                  | Septic equalization tank, anaerobic zones, sludge storage             | Increase aeration/mixing, cover tanks with biofilter/scrubber, timely sludge removal                    |
| Low BOD/COD removal efficiency                     | Under-sized aeration, toxic shock load, inadequate nutrients          | Equalize shock loads, verify C:N:P ratio (100:5:1), check DO and MLSS, inspect for toxic inhibition     |
| High effluent color (textile/dye units)            | Non-biodegradable dye molecules                                       | Add coagulant-assisted color removal, ozone/AOP polishing, activated carbon adsorption                  |
| Membrane/RO fouling in tertiary reuse              | Colloidal fouling, scaling, biofouling                                | Pretreat with UF/cartridge filters, antiscalant dosing, periodic CIP (clean-in-place)                   |
| Frequent pump/blower failure                       | Poor pretreatment (rags/grit ingress), lack of preventive maintenance | Upgrade screening, implement PM schedule, install standby units                                         |
| High sludge generation cost                        | Over-dosing of coagulants, poor dewatering                            | Optimize jar-test based dosing, upgrade dewatering equipment, consider sludge minimization technologies |
| Non-compliance with discharge norms                | Design capacity exceeded, equipment ageing, operator error            | Conduct capacity audit/revamp, retrain O&M staff, install online analyzers with SCADA alarms            |

9. Equipment Service Life

| Equipment                                | Typical Service Life                          | Key Life-limiting Factors                                  |
|------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
| RCC/Steel Tanks & Civil Structure        | 25-30 years                                   | Chemical attack, waterproofing quality, structural loading |
| Screens (mechanical/static)              | 8-12 years                                    | Corrosion, abrasive solids, mechanical wear                |
| Submersible/Centrifugal Pumps            | 5-8 years (impeller/seals 2-3 yrs)            | Cavitation, solids abrasion, seal wear, corrosive media    |
| Aeration Blowers (roots/centrifugal)     | 10-15 years (rebuild every 4-5 yrs)           | Bearing wear, filter maintenance, duty cycling             |
| Diffusers (fine bubble membrane)         | 5-7 years                                     | Membrane fouling/stiffening, biofilm clogging              |
| Agitators/Flocculators (gearbox & motor) | 10-12 years                                   | Gearbox oil quality, shaft alignment, corrosion            |
| Filter Press Plates/Cloths               | Plates 10+ yrs; cloths 6-18 months            | Chemical/abrasive wear, cake release efficiency            |
| Decanter Centrifuge                      | 10-15 years (bowl/scroll overhaul at 5-7 yrs) | Abrasive solids, vibration, bearing wear                   |
| Instrumentation (pH/DO/level probes)     | 2-5 years                                     | Membrane fouling, calibration drift, chemical exposure     |
| Dosing Pumps                             | 5-7 years                                     | Diaphragm wear, chemical compatibility, priming issues     |
| MBBR Media Carriers                      | 15-20 years                                   | UV degradation (if exposed), mechanical abrasion           |
| RO Membranes (if tertiary reuse)         | 3-5 years                                     | Fouling, scaling, oxidant damage, poor pretreatment        |

10. Preventive Service & Maintenance Schedule

| Frequency   | Activity                                                                                                                                                                                             |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Daily       | Visual inspection of screens for choking; check pH, DO, MLSS visually; log flow meters; check for foam/odor/color abnormality; verify dosing pump operation; check sludge blanket depth in clarifier |
| Weekly      | MLSS/MLVSS lab test; SVI (Sludge Volume Index) test; grease trap cleaning; lubrication check on rotating equipment; inspect diffusers for even air distribution                                      |
| Monthly     | Full lab characterization of inlet/outlet (BOD, COD, TSS, oil & grease); calibrate online analyzers; inspect blower belts/couplings; check dosing pump calibration                                   |
| Quarterly   | Vibration analysis on pumps/blowers; gearbox oil sampling; clean/inspect diffuser membranes; inspect filter press cloths for blinding; review sludge disposal manifest & compliance                  |
| Half-Yearly | Overhaul dosing systems; inspect civil structure for cracks/corrosion; test emergency power/standby pump changeover; internal audit of SOPs                                                          |
| Annual      | Complete shutdown & internal inspection of tanks/clarifiers; blower/pump major overhaul as per OEM schedule; recalibrate all                                                                         |

## 11. Troubleshooting Guide

| Symptom                                | Probable Cause                                               | Corrective Action                                                                      |
|----------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Treated water turbid/high TSS          | Clarifier hydraulic overload, weak floc, short-circuiting    | Check coagulant dose via jar test, reduce flow, inspect clarifier weir levelness       |
| Aeration tank DO consistently low      | Blower under-capacity, diffuser fouling, organic overload    | Clean/replace diffusers, verify blower output, check for shock load source             |
| Sudden drop in pH treatment efficiency | Dosing pump failure, chemical stock-out, probe drift         | Verify dosing pump calibration, check chemical inventory, recalibrate/replace pH probe |
| Excess sludge floating in clarifier    | Denitrification (rising sludge), long sludge retention       | Increase sludge wasting rate, reduce SRT, improve RAS/WAS control                      |
| Discharge exceeding COD norms          | Toxic/organic shock load, insufficient HRT, bio-mass die-off | Trace-back source, equalize load, reseed biomass, consider AOP polishing               |
| Pump not delivering rated flow         | Clogged strainer, worn impeller, air lock, wrong rotation    | Clear strainer, inspect impeller wear ring, bleed air, verify motor rotation direction |
| Frequent MCC/motor tripping            | Overload from solids ingress, single-phasing, poor earthing  | Improve pretreatment screening, check phase balance, verify earth pit resistance       |
| Odor complaint from surrounding area   | Anaerobic conditions in equalization/sludge storage          | Increase aeration, install biofilter/carbon scrubber, reduce sludge holding time       |
| High energy consumption                | Over-aeration, blower inefficiency, pump oversizing          | Install DO-based blower VFD control, conduct energy audit, right-size pumps            |

## 12. Project Execution & Engineering Workflow

| Phase                                 | Key Deliverables / Activities                                                                                                                                                                  |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feasibility & Effluent Survey         | Water balance, effluent characterization (min. 3-6 months data), site survey, regulatory norm study, technology screening, budgetary cost estimate.                                            |
| Basic Engineering (Concept/FEED)      | Process Flow Diagram (PFD), mass balance, preliminary P&ID, equipment sizing, layout concept, capex estimate ( $\pm 20\text{-}30\%$ accuracy).                                                 |
| Detailed Engineering                  | Final P&IDs, hydraulic profile/GA drawings, structural & civil drawings, electrical single line diagram, instrumentation & control philosophy, equipment datasheets, tender/BOQ documentation. |
| Statutory Approvals                   | Consent to Establish (CTE) from SPCB, environmental clearance if applicable, local body NOCs, fire & electrical inspectorate approvals.                                                        |
| Procurement                           | Vendor evaluation, technical bid evaluation, purchase orders for major equipment, expediting & inspection.                                                                                     |
| Civil & Structural Construction       | Tank construction (RCC/steel), foundation for equipment, piping trenches, plant roads and drainage.                                                                                            |
| MEI Erection                          | Equipment installation and alignment, piping and valve installation, cabling, panel installation, SCADA/PLC integration.                                                                       |
| Pre-commissioning                     | Dry run of all rotating equipment, loop checking, hydro-testing of tanks/pipelines, calibration of instruments, safety interlocks testing.                                                     |
| Commissioning & Biomass Seeding       | Wet trial with actual/simulated effluent, biological seeding and acclimatization (2-6 weeks), fine-tuning of dosing and aeration.                                                              |
| Performance Guarantee Test Run (PGTR) | Continuous run (typically 72 hours to 30 days) to demonstrate contractual outlet quality at rated design flow and load.                                                                        |
| Handover & O&M                        | As-built drawings, O&M manuals, spares philosophy, operator training, warranty/DLP support, CTO renewal support.                                                                               |

## 13. Statutory Compliance & Performance Guarantee

### 13.1 Typical Discharge Norms (Inland Surface Water, illustrative)

| Parameter | Typical Permissible Limit |
|-----------|---------------------------|
|           |                           |

|                    |                                   |
|--------------------|-----------------------------------|
| pH                 | 5.5 - 9.0                         |
| BOD (3 days, 27°C) | ≤ 30 mg/L                         |
| COD                | ≤ 250 mg/L                        |
| TSS                | ≤ 100 mg/L                        |
| Oil & Grease       | ≤ 10 mg/L                         |
| Total Chromium     | ≤ 2 mg/L                          |
| Color              | As per CPCB/SPCB specific consent |

Actual limits are notified in the plant-specific Consent to Operate (CTO) and may be more stringent based on receiving water body classification and industry category.

### 13.2 Performance Guarantee Test Run (PGTR) Essentials

- Conducted at guaranteed design flow (or agreed minimum, typically 80% of design capacity).
- Third-party/NABL-accredited lab used for inlet-outlet sample analysis.
- Continuous run duration and sampling frequency defined contractually (commonly 72 hours to 30 days).
- Deviation/rejection clause and liquidated damages defined in the contract for non-achievement.
- Successful PGTR triggers final payment milestone and start of Defect Liability Period (DLP), typically 12 months.

Closing note: a well-engineered ETP is only as reliable as its pretreatment design and its preventive maintenance discipline. The majority of compliance failures in operating plants trace back to inadequate pretreatment (solids/oil ingress), under-sized equalization, or deferred maintenance rather than to the core biological/chemical process design. Periodic capacity audits against actual (not nameplate) production levels are strongly recommended every 3-5 years.